

UDAIPUR, India

WMO: 425420

Lat: 24.618N

Lon: 73.896E

Elev: 513

StdP: 95.31

Time Zone: 5.50 (IND)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 7.1	(c) 8.6	(d) -3.1	(e) 3.1	(f) 22.2	(g) -1.1	(h) 3.7	(i) 23.2	(j) 5.1	(k) 22.0	(l) 4.4	(m) 21.8	(n) 0.6	(o) 320	(p) 0.334

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	11.6	41.1	20.8	40.0	20.5	38.8	20.4	26.6	31.5	26.1	30.4	25.7	29.8	3.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.6	22.2	28.3	25.1	21.6	27.8	24.8	21.1	27.5	87.4	31.8	85.0	30.0	83.0	29.9	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
5.5	4.8	4.2		4.6	43.0	1.5	1.6	3.5	44.2	2.6	45.2	1.8	46.1	0.7	47.3	
			DB	2.6	28.6	1.1	1.3	1.8	29.6	1.2	30.3	0.6	31.1	-0.1	32.0	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.2	16.1	19.7	24.8	30.1	33.2	32.0	28.4	27.0	27.1	25.6	21.4	17.2
	DBStd	5.76	2.29	2.72	3.01	2.42	1.99	2.46	1.89	1.09	1.40	1.78	2.06	2.31
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	142	76	16	1	0	0	0	0	0	0	0	2	49
	CDD10.0	5561	190	272	460	602	720	659	570	526	512	484	342	224
	CDD18.3	2662	7	54	202	352	462	409	312	267	262	226	93	15
	CDH23.3	36088	159	654	2681	5531	7770	6188	3703	2610	2777	2688	1088	239
	CDH26.7	19266	19	200	1399	3524	5342	3843	1505	693	1078	1306	329	27
Wind	WSAvg	1.6	1.1	1.3	1.5	1.9	2.4	2.5	2.1	1.7	1.4	1.0	0.9	1.0
Precipitation	PrecAvg	636	3	2	4	7	8	60	241	154	116	12	3	0
	PrecMax	1027	35	12	30	39	43	203	453	401	281	77	79	10
	PrecMin	361	0	0	2	0	0	0	82	4	12	0	0	0
	PrecStd	173	8	3	6	11	13	39	89	102	79	21	15	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.9	32.7	38.6	41.5	43.2	43.0	37.0	32.8	35.3	35.2	32.1	29.4
		MCWB	16.3	16.9	18.4	18.8	20.4	23.6	23.8	24.9	23.2	20.0	17.5	17.0
	2%	DB	26.8	30.9	36.4	39.9	41.2	40.9	34.4	31.6	34.0	34.1	30.9	27.2
		MCWB	15.3	16.8	19.5	18.7	20.8	22.5	24.6	25.0	23.2	20.0	18.2	16.2
	5%	DB	25.1	29.3	34.9	38.8	40.3	38.8	33.1	30.8	32.8	33.2	29.8	25.9
		MCWB	14.5	16.3	18.6	18.7	20.5	23.1	24.7	24.8	23.1	20.0	17.8	15.5
	10%	DB	23.4	27.5	33.3	37.7	39.2	36.9	32.0	29.9	31.6	32.3	28.7	24.6
		MCWB	13.8	15.7	17.4	18.8	20.7	23.7	24.5	24.6	23.3	19.9	17.4	15.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.5	19.8	24.9	23.8	26.3	28.3	26.9	26.7	26.6	24.9	22.5	19.4
		MCDB	25.6	27.6	35.5	35.0	35.5	36.7	31.1	30.0	30.2	29.5	26.6	25.9
	2%	WB	16.3	17.7	21.7	21.5	25.0	26.9	26.3	26.1	25.8	23.9	21.3	17.4
		MCDB	23.8	27.7	32.1	34.2	35.1	33.3	30.0	29.1	29.4	28.9	25.5	25.1
	5%	WB	15.3	17.1	18.9	20.3	24.0	26.0	25.8	25.7	25.3	23.1	19.9	16.4
		MCDB	23.1	27.3	32.0	33.8	34.6	32.0	29.4	28.5	28.7	28.3	25.7	23.9
	10%	WB	14.4	16.3	17.9	19.7	23.2	25.4	25.5	25.3	24.8	22.2	18.8	15.6
		MCDB	22.1	26.1	31.0	34.0	34.4	31.8	29.0	28.0	28.3	28.2	26.2	23.0
Mean Daily Temperature Range	5% DB	MDBR	13.8	14.3	14.8	14.1	11.6	8.7	5.5	5.2	7.6	12.8	13.3	13.6
		MCDBR	16.6	16.6	16.5	15.3	12.6	11.2	7.7	7.1	10.5	14.1	15.1	15.3
	5% WB	MCWBR	7.2	6.3	5.8	4.2	3.3	3.6	1.9	2.3	3.2	4.0	5.3	6.5
		MCWBR	14.5	14.6	14.8	13.0	10.9	8.7	5.7	5.4	6.7	10.0	10.8	13.3
Clear-Sky Solar Irradiance	taub		0.388	0.394	0.442	0.499	0.548	0.638	0.663	0.601	0.531	0.478	0.473	0.420
		taud	2.187	2.198	2.067	1.960	1.884	1.742	1.718	1.864	2.002	2.031	1.978	2.086
	Ebn at Noon		857	883	857	815	771	699	681	725	768	788	758	801
		Edh at Noon	130	138	165	188	203	232	237	205	174	161	160	139
All-Sky Solar Radiation	RadAvg		4.31	5.27	6.34	7.12	7.35	6.27	4.49	4.24	5.17	5.47	4.50	4.08
	RadStd		0.20	0.14	0.22	0.19	0.16	0.42	0.48	0.43	0.47	0.21	0.27	0.18

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a) Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(b) Regional (0 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A

Nomenclature: See separate page